

~~DvGfhdud hvGdvGluh hvGdGdGhu rGhvs hf lydp hq hC~~

~~$$d = \frac{x^y}{w} y^z C$$~~

C

~~$$d = \frac{z}{w} x^z y^z z^z C$$~~

C

~~dvrGdvfrdp hq rGhndG hup dq hq hChp vhc~~

~~$$G^x_w . G^y_w G^z_w$$~~

Forças atuantes em um elemento fluido

$$\frac{p}{w} \quad G_o$$

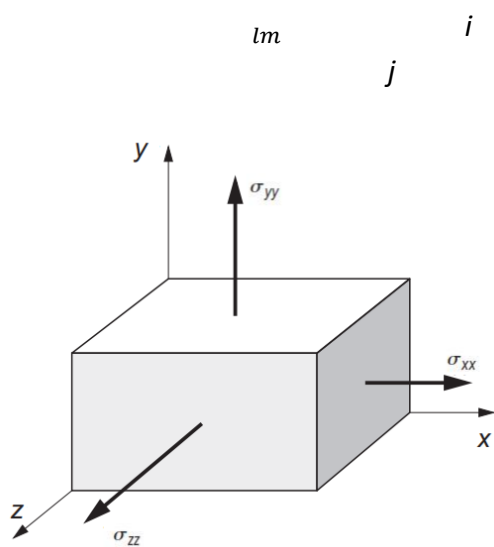
$$\frac{p}{ro} \quad \tau$$

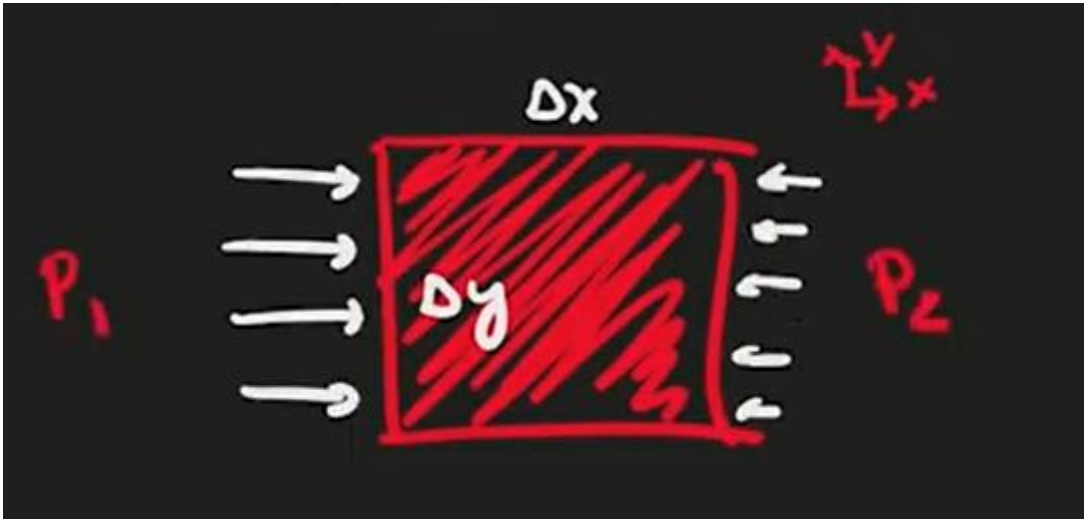
$$\frac{\tau}{w} \quad G_o$$

$$\tau \quad \frac{G}{w} \quad o$$



As forças normais ou de pressão





$$s \quad \frac{G}{D} \quad G D$$

$$s \ D \quad s \ D$$

$$D \quad D$$

$$s \quad s$$

$$s \quad s \quad s$$

$$s$$

$$\frac{\tau}{r_0} = \frac{S}{r_0}$$

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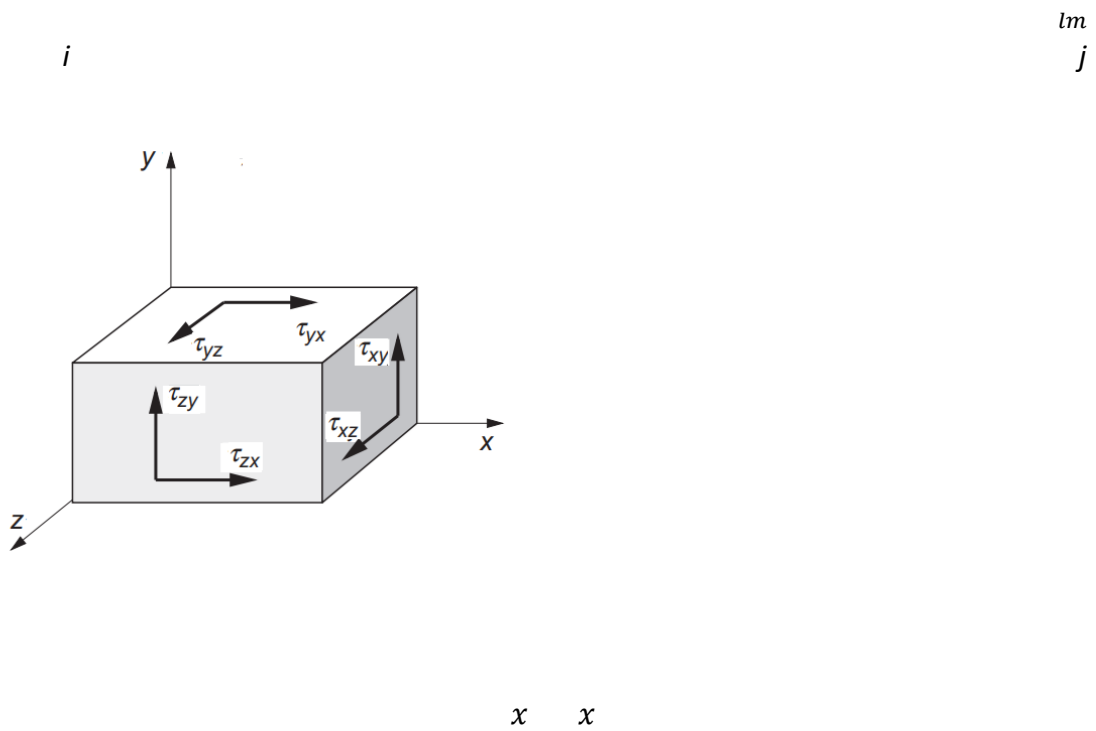
$$\frac{\tau}{r_0} = \frac{S}{r_0}$$

$$\tau \frac{x}{w} = \frac{S}{w}$$

$$\tau \frac{y}{w} = \frac{S}{w}$$

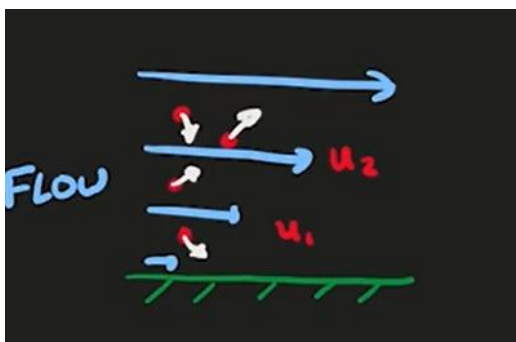
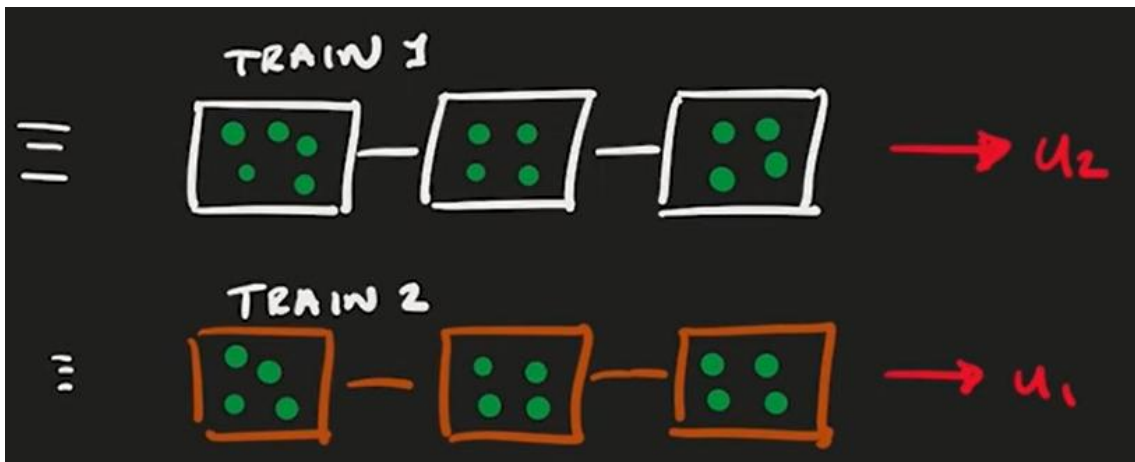
$$\tau \frac{z}{w} = \frac{S}{w}$$

As forças tangenciais, também conhecidas como forças friccionais ou viscosas



x

x



C_D

C_D

$$C_D = \frac{p}{D} \frac{x}{w}$$

$$\frac{x}{w}$$

—

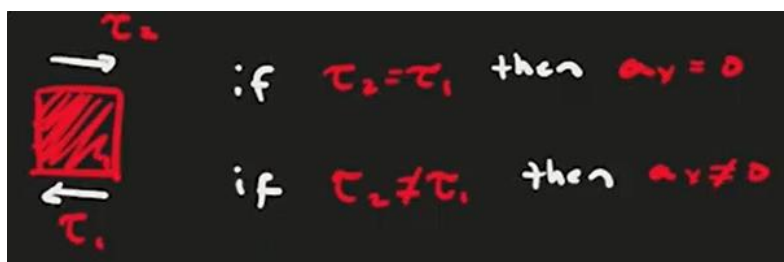
$$\frac{x}{w} = \frac{x}{w} = \frac{x}{w}$$

$$\frac{p}{D} \frac{x}{w} = \frac{p}{D} \frac{x}{w}$$

$$\frac{p}{D} \frac{x}{w}$$

$$\frac{p}{D} \frac{x}{w}$$

$$\frac{x}{w}$$





D *D*

D *D*

x

x

x

o

x

o

x

x

x

$\tau \frac{x}{w}$

s

x

x

x

$$\tau \frac{y}{w} \quad \frac{s}{\quad} \quad \frac{y}{\quad} \quad \frac{y}{\quad} \quad \frac{y}{\quad}$$

$$\tau \frac{z}{w} \quad \frac{s}{\quad} \quad \frac{z}{\quad} \quad \frac{z}{\quad} \quad \frac{z}{\quad}$$

~~As forças de corpo~~

~~ϕ ϕ~~

~~$\frac{\quad}{\quad}$ ϕ~~

$$\tau \frac{x}{w} \quad \frac{s}{\quad} \quad \frac{x}{\quad} \quad \frac{x}{\quad} \quad \frac{x}{\quad} \quad \tau$$

$$\tau \frac{x}{w} \quad x \frac{x}{\quad} \quad y \frac{x}{\quad} \quad z \frac{x}{\quad} \quad \frac{s}{\quad} \quad \frac{x}{\quad} \quad \frac{x}{\quad} \quad \frac{x}{\quad} \quad \tau \quad C$$

~~hGr up dGlp lduGdlGluh hvôôôôGhu rGhvs hf lydp hq hC~~

$$\tau \frac{\quad}{w} \quad x \frac{y}{\quad} \quad y \frac{y}{\quad} \quad z \frac{y}{\quad} \quad \frac{s}{\quad} \quad \frac{y}{\quad} \quad \frac{y}{\quad} \quad \frac{y}{\quad} \quad \tau \quad C$$

C

$$\tau \frac{z}{w} \quad x \frac{z}{\quad} \quad y \frac{z}{\quad} \quad z \frac{z}{\quad} \quad \frac{s}{\quad} \quad \frac{z}{\quad} \quad \frac{z}{\quad} \quad \frac{z}{\quad} \quad \tau \quad C$$

C